Design and Analysis of Algorithms (CSC-316) Instructor: **Muhammad Zaman Aziz**
Credit Hrs: **3** Semester: **Fall 2025**
Pre-requisite: **Data Structures and Algorithms (CSC-214)**
Contact Hrs:

Course Objective:

The aim of this course is to teach students to reason about algorithms and their time and space requirements. In addition, many known algorithms to solve known problems will be discussed. At the end of the course, students should be able to analyze the efficiency of different types of algorithms.

The main concepts discussed are: asymptotic notations, recursion and recurrence relations, divide-and-conquer approach, sorting, search trees, heaps, greedy approach, dynamic programming, graph algorithms and shortest paths.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- PLO5: Modern Tool Usage
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PL10: Life-long Learning

Course Learning Outcomes (CLOs):

CLO No.	CLO Statement	Bloom Taxonomy	PLOs/ Graduate Attributes (Seoul Accord)
CLO1	Describe the various algorithm design paradigms, including divide and conquer, dynamic programming, and greedy algorithms, and explain their suitability for solving different computational problems.	C2	PLO2
CLO2	Explain the major string matching and graph algorithms and their analysis.	C2	PLO2
CLO3	Analyze an algorithm's performance using asymptotic analysis.	C4	PLO3
CLO4	Choose an appropriate algorithm from a set of algorithms for a given problem.	C5	PLO4

	PLO2	PLO3	PLO4
CLO1	✔		
CLO2	✔		
CLO3		✔	
CLO4			✔

Text Book:
(A) Introduction to Algorithms (4th edition,2022), Thomas H. Corman, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein, MIT Press

Reference Book:
(B) Algorithm Design (1st edition, 2005), Jon Kleinberg and Eva Tardos, Pearson
(C) Algorithms (4th edition, 2011), Robert Sedgewick, Kevin Wayne, Addison-Wesley Professional
(D) Foundations of Algorithms (5th edition,2014), Richard E Neopolitan, Jones & Bartlett Learning

Course Outline and Schedule

Week Mon-Sat	Topics	Text Book	CLO	PLO
Wk-1 22-Sep	The role of algorithms in computing Asymptotic functions and notations Analysis of simple algorithms	Chap 1&3	CLO3	PLO3
Wk-2 29-Sep	**Sorting Algorithms and their analysis** • Comparison Based Sorting Algorithms: • Selection, Bubble, Insertion Sort • Lower Bound on Comparison based sorting • Linear Time Sorting (Non-Comparison Based Sorting • Counting, Radix Sort, Bucket Sort **Loop Invariants**	Chap 2 & 8	CLO3 CLO4	PLO3 PLO4
Wk-3 6-Oct	**Divide and Conquer** • Merge Sort • Analysis of Merge Sort	Chap 2	CLO1 CLO3	PLO2 PLO3
Wk-4 13-Oct	**Recursion and recurrence relations** • Recursion Tree Method • Master Theorem	Chap 4	CLO4	PLO4
Wk-5 20-Oct	**Divide and Conquer** • Quick Sort • Analysis of Quick Sort	Chap 7	CLO1 CLO3	PLO2 PLO3

Wk-6 27-Oct	• Variations of Quick Sort	Chap 7	CLO1 CLO4	PLO2 PLO4
Wk-07 3-Nov	**Search Trees** • Binary Search Trees • Balanced Search Trees	Chap 12	CLO3 CLO4	PLO3 PLO4
Wk-8 10-Nov	**Heaps** **Hashing**	Chap 6	CLO3 CLO4	PLO3 PLO4
Wk-9 17-Nov	**Mid Term Exam**			
Wk-10 24-Nov	**Greedy Algorithms** • Activity selection Problem • Huffman Coding	Chap 16	CLO1 CLO3 CLO4	PLO2 PLO3 PLO4
Wk-11 1-Dec	**Dynamic Programming** • Assembly Line Scheduling • 0/1 Knapsack • Matrix Chain Multiplication	Chap 15	CLO1 CLO3 CLO4	PLO2 PLO3 PLO4
Wk-12 8-Dec	**Graph Algorithms** General Properties Elementary Graph Algorithms • BFS • DFS	Chap 22	CLO2 CLO3	PLO2 PLO3
Wk-13 15-Dec	Minimum Spanning Trees • Prim's algorithm • Kruskal's algorithm	Chap 23	CLO2 CLO3	PLO2 PLO3
Wk-14 22-Dec	Single-Source Shortest Paths • Dijkstra's algorithm • Bellman-Ford algorithm All-Pairs Shortest Paths • Floyd-Warshall algorithm	Chap 24 &25	CLO2 CLO3 CLO4	PLO2 PLO3 PLO4
Wk-15 29-Dec	**String Matching** The Boyer-Moore Algorithm The Rabin-Karp algorithm The Knuth-Morris-Pratt algorithm **NP completeness**	Chap 32	CLO2 CLO3 CLO4	PLO2 PLO3 PLO4
Wk-16 5-Jan	Presentations			
10- Jan to 13- Jan	**Reinforcement (Exam Prep)**			
14-Jan to 24- Jan	**Final Exam**			
29-Jan	**Exam Scripts Shown**			

30-Jan	**Declaration of Result**			

Grading Breakdown and Policy

ASSESSMENT DESCRIPTION	WEIGHT (%)	CLOS	PLOS
Quizzes / Class Tests (3-4)	12%	CLO1- CLO4	PLO2, PLO3, PLO4
Assignments (3-4)	12%	CLO1, CLO2, CLO3	PLO2, PLO3
Presentation	6%	CLO3	PLO3
Mid Term Exam	25%	CLO1, CLO3, CLO4	PLO2, PLO3, PLO4
Final Exam	45%	CLO1- CLO4	PLO2, PLO3, PLO4

GRADING TABLE (Absolute Scale given below is applicable for up to 20 students in Final Exam; otherwise Relative Grading is applicable)			
Range	**Percentage**	**Points**	**Grade**
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A-
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B-
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C-
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F